

COMAP AI Use Disclosure Report

HiMCM 2022 Problem A — The Need for Bees

Team XXXX

This report documents artificial-intelligence assistance used while preparing our contest materials, in accordance with COMAP contest rules on AI disclosure. Tooling was limited to the Cursor IDE agent for code scaffolding and LaTeX formatting support. No AI system was used to invent numerical results; all tables and figures are produced by executing our Python pipeline and reading `results/*.csv`.

1 Tools and scope

Tool	Role	Contested artifacts touched
Cursor (AI coding agent)	Assisted PyTorch training-loop / Euler unroll drafting	<code>src/autograd_engine/tor</code>
Cursor	Assisted elasticity & bifurcation scripts	<code>src/sensitivity/</code> , <code>src/st</code>
Cursor	Assisted orchard OR module & pytest path config	<code>src/pollination/orchard</code>
Cursor	Assisted LaTeX table/figure wiring & document shells	<code>paper/*.tex</code>

Interaction prompts that affected model or algorithm design were logged in the pack file `prompts/cursor_log.md` for auditability.

2 What the human team did independently

We explicitly attest that the following were conceived, specified, and signed off by the student team without outsourcing scientific judgment to an AI system:

1. **Differential-equation mechanism:** choice of the five compartments B, H, F, P, N , dual-feedback structure (social inhibition and nutritional gating), and stress-channel meanings (pesticide, Varroa, frost).
2. **Activation-function smoothing design:** decision to use Holling-III, Softplus state projection, and seasonal $\Omega(t)$ so that Autograd elasticities are well-defined.
3. **Summary Sheet:** narrative priorities and the final one-page claims; every numeric entry was typed from measured CSV outputs after human inspection.
4. **Grower / association advisory letter:** field recommendations on fan placement, bloom feeding, and Varroa triggers were written as a human policy brief; AI only helped with LaTeX layout.

3 What AI assistance was used for

- Drafting boilerplate for PyTorch `nn.Module` parameter registration and day-loop bookkeeping.
- Generating initial LaTeX `booktabs` tables and `includegraphics` paths tied to existing figure filenames.

- Fixing pytest `pythonpath` so `from src... imports resolve`.

AI suggestions were reviewed, edited, and re-run by the team. When an AI-proposed number disagreed with a CSV, the CSV prevailed.

4 Data integrity statement

We did not ask any AI system to fabricate experimental outcomes. Peak populations, elasticities, CCD tips ($\mu_F^{*,\text{soft}} = 0.11$, $\mu_F^* = 0.305$), and orchard optima ($K^* = 40$, YieldRatio 0.848) are copied from

```
ccd_healthy_vs_stress_snapshot.csv,  
autograd_elasticity_ranking.csv,  
ccd_tipping_point.csv,  
orchard_pollination_optimization.csv.
```

5 Signatures

Team member names / roles: _____

Date: _____

Contest control number: XXXX